

NAME: MOHAMMED ALMAS

ROLL NO: A24126552098

BRANCH: CSM

SECTION: B

COURSE NAME: FULL STACK WEB DEVELOPMENT

CODE: 23CSM4121

TITLE OF THE EXPERIMENT: Dynamic To-Do List using JavaScript and DOM Manipulation

DESCRIPTION:

The To-Do List application is a dynamic and interactive webpage developed using **HTML, CSS, and JavaScript**. It allows the user to enter a task in a text box and add it to the task list by clicking the **Add Task** button.

When a task is added, JavaScript dynamically creates a new task item along with **Complete** and **Delete** buttons using DOM manipulation. The **Complete** button visually marks the task as completed, while the **Delete** button removes the selected task from the webpage.

The application also displays "**No tasks available.**" when the task list is empty. This program demonstrates **DOM element selection, dynamic element creation, event listeners, click event handling, adding and removing elements, and interactive webpage development.**

PROGRAM:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>My To-Do List</title>
```

```
  <style>
```

```
body {  
    font-family: Arial;  
    background-color: #f2f2f2;  
    text-align: center;  
    padding: 30px;  
}
```

```
input {  
    padding: 10px;  
    width: 250px;  
}
```

```
button {  
    padding: 10px 15px;  
    margin: 5px;  
    cursor: pointer;  
}
```

```
#taskList {  
    width: 400px;  
    margin: 20px auto;  
}
```

```
.task {  
    background-color: white;  
    padding: 10px;
```

```
margin: 10px;
border-radius: 5px;
display: flex;
justify-content: space-between;
align-items: center;
}
```

```
.completed {
  text-decoration: line-through;
  color: gray;
}
```

```
#emptyMessage {
  color: gray;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>My To-Do List</h1>
```

```
<input type="text" id="taskInput" placeholder="Enter a task...">
```

```
<button id="addButton">Add Task</button>
```

```
<div id="taskList">
```

```
<p id="emptyMessage">No tasks available.</p>
```

```
</div>
```

```
<script>
```

```
let taskInput = document.getElementById("taskInput");
```

```
let addButton = document.getElementById("addButton");
```

```
let taskList = document.getElementById("taskList");
```

```
let emptyMessage = document.getElementById("emptyMessage");
```

```
addButton.addEventListener("click", function() {
```

```
    let taskText = taskInput.value;
```

```
    if (taskText.trim() === "") {
```

```
        return;
```

```
    }
```

```
    // Remove empty message
```

```
    emptyMessage.style.display = "none";
```

```
    // Create task container
```

```
    let task = document.createElement("div");
```

```
task.className = "task";

// Create task text
let text = document.createElement("span");
text.textContent = taskText;

// Create Complete button
let completeButton = document.createElement("button");
completeButton.textContent = "Complete";

// Complete task
completeButton.addEventListener("click", function() {
    text.classList.toggle("completed");
});

// Create Delete button
let deleteButton = document.createElement("button");
deleteButton.textContent = "Delete";

// Delete task
deleteButton.addEventListener("click", function() {

    task.remove();

// Show message if no tasks remain
    if (taskList.querySelectorAll(".task").length === 0) {
```

```
        emptyMessage.style.display = "block";
    }
});

// Add elements to task
task.appendChild(text);
task.appendChild(completeButton);
task.appendChild(deleteButton);

// Add task to list
taskList.appendChild(task);

// Clear input
taskInput.value = "";
});
```

```
</script>
```

```
</body>
```

```
</html>
```

OUTPUT:

My To-Do List

Complete Assignment

Complete

Delete

Prepare for Quiz

Complete

Delete

Practice JavaScript

Complete

Delete

